

Cody Lee

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Building end-to-end LLM pipelines — retrieval, agent loops, evaluation, and cost-tiered routing — and forensically documenting where they break across legal and financial documents, trading systems, and production stacks.

Education

University of California, Riverside • B.S. Computer Science

Expected Jun 2026

Relevant coursework: Natural Language Processing, Artificial Intelligence, Data Science, Probability & Statistics

Projects

Meridian — Forensic RAG Measurement Framework [8-page research](#)

May 2026

- Built a **two-layer measurement framework** for retrieval-augmented generation, validated across the **LegalBench-RAG** legal benchmark (4 corpora — contracts, M&A agreements, NDAs, privacy policies; **~800 queries** over **~79M characters / ~20M tokens**).
- **Layer 1** — a deterministic span-overlap failure taxonomy (six failure types, **no LLM judge**) calibrated to the published **Ze-roEntropy** baseline; the optimized system reached **retrieval Precision@1 2–20×** that baseline (full stack vs. their dense-only baseline). Retrieval stack also transferred zero-shot to **NFCorpus** (medical IR, **3,633 docs**), beating classic BEIR baselines.
- **Layer 2** — pinned-judge answer correctness + groundedness, fully isolated from Layer 1 so a metric change is never mistaken for a pipeline change. A config stack (**summary-augmented chunking + convex-combination fusion + document routing + LLM selector**) lifted end-to-end answer correctness **+7.9pp** average across four corpora — up to **25.8%→75.3%** on contracts.

Aether — LLM Workflow Engine

Apr 2026

- Built a **reason-act-observe agent** that turns a financial document and a plain-language goal into a **grounded chart** where every value traces to a source cell — direct SDK, no LangChain.
- **75.5% end-to-end on FinQA** (financial QA benchmark, n=200); retrieval R@5 0.85, MRR@3 0.73 over 200 queries; six-tool loop with per-step Pydantic validation.
- Aggregated a **700-row financial dataset** into per-segment SQL totals matching an independent computation exactly; full **observability** — every step and tool call traced to SQLite and rendered as a visual reasoning trace.

Polymarket Trading Bot Autopsy [2-page summary](#) | [15-page autopsy](#)

Apr–Jun 2026

- Found **three measurement bugs** that inflated paper trading results **~135×** and made the strategy misleading — the bots flagged as best performed worst live.
- Built a **3-layer LLM classification pipeline** (Haiku → Sonnet → Opus) over 2,409 wallets; 180 bots, 417K simulated trades. Tiered routing cut cost to \$30–40 vs **~\$150** on Opus alone.

ChainTax — Crypto Tax Engine

Feb–Mar 2026

- Built a **cross-source ingestion and reconciliation pipeline**: 540K+ events across Hyperliquid and two on-chain APIs (Alchemy, DeBank), with bridge detection to prevent double-counting across chains.
- **FIFO lot matching** with three competing IRS funding treatments producing three Form 8949 variants per run, plus offsetting-position detection across spot, perp, and correlated pairs.

Aegis — Verifiable RL Environment for Defensive Cybersecurity

In progress

- Building a **reinforcement-learning environment** where an agent patches real vulnerabilities in sandboxed repositories, scored by a **multi-signal deterministic verifier** engineered to resist **reward hacking** — extending Meridian's deterministic-measurement approach from retrieval metrics to RL reward integrity.
- Designing a **two-track evaluation**: agent capability benchmarked against standardized vuln-repair suites (ZeroDayBench, SecureAgentBench), and **verifier integrity scored as a classifier** against a labeled set of genuine-vs-reward-hacked patches — measuring whether **reward hacking rises under optimization pressure**.

Technical Skills

Languages Python, TypeScript, SQL, Bash

LLM / RAG Hybrid retrieval (dense + sparse), chunking, embedding, reranking, fusion (RRF / convex-combination), deterministic span-overlap evaluation, vector DBs (Qdrant, Pinecone), LangGraph, MCP servers, Anthropic / OpenAI / Google SDKs

Data & Observability Cross-source ingestion, pandas, DuckDB, SQLite; Langfuse tracing, cost / latency

Infrastructure Git, REST APIs, FastAPI, Streamlit, VPS deployment

Methodology Forensic measurement, failure-mode analysis, statistical evaluation, technical documentation